Exam B — 036026 Kinematics, Dynamics and Control of Robots

Spring Semester 2025

Exam information and instructions

- Duration: 09:00 - 12:00, July 11, 2025
- You are allowed to use only the lab computer at the workstation allocated to you. You are not allowed to use your own laptop or a tablet (like an iPad).
- The first 30 minutes of the exam is closed book, where you need to answer question 1 on Moodle, using the Safe Exam Browser. You are allowed to use a calculator to answer question 1. Question 1 consists of five short questions counting 3 marks each. You have only one opportunity to submit your answers; no repeated submissions are allowed. After 30 minutes (at 09:30) question 1 closes on Moodle and you will no longer have access to it. Make sure you submit your answers before it closes.
- After question 1 closes, the exam proceed to an open book part for the next 150 minutes, during which you are allowed to use paper notes and books. You are allowed to load data on the computer from a USB flash drive or external drive. You are also allowed to download files from your own space on Moodle. You may start with the open book part of the exam during the first 30 minutes of the exam, but you are not allowed to use your notes or books during the first 30 minutes.
- You are not allowed to use the internet except for uploading files or information to or from Moodle. You are not allowed to communicate with another person via email, WeChat or whatever other means possible. If you are caught using the internet — except Moodle — or communicating with a person other than the invigilators or the lecturer during the exam, this will be considered as academic dishonesty and disciplinary action against you will follow.
- You need to download various files that you need to use in answering questions 2, 3 and 4, from Moodle. All these files are in the zip file “Exam_B_files.zip”, which is available on Moodle under the “Exams” topic.
- The answers you present to the open book part of the exam should show a logical train of thought, starting from the theory that was covered in the reading assignments. If you use your computer to solve your problems, remember that the grader of your exam does not see your computer script and output. You must therefore supply enough written information in your exam script to allow the grader to assess your answer. If you solve equations symbolically or numerically in MATLAB or Python, these equations must be written in your exam script, with an appropriate and logical explanation of how they were derived.
- In final answers to questions, all fractions, products, sums, exponential expressions and natural constants like π should be evaluated to single real numbers, given accurately to four significant digits.
- For questions 2 and 3 you need to fill in variables or parameters that you have calculated in a feedback tool on Moodle. All the variables/parameters that you fill in in the feedback

tool, you must also clearly write in your exam script, so that it can be confirmed after the exam that the values entered on Moodle are actually the values that **you** have determined in the exam. If you fail to supply these parameters in your script, your answers on Moodle may be ignored and you will either be severely penalized or get zero for the corresponding part of the question. If you write the parameters in your script only, you will certainly loose marks.

- In all questions use $g = 9.81$ m/s^2 where required.
- The notation used in this exam is the same as what is used in the course text book by Murray, Li and Sastry, and in the course presentation, unless indicated otherwise.
- The paper consists of four questions. Full marks are 100.

Question 1 (15 marks)

Question 1 is asked as a quiz on Moodle, under the "Exam" topic. This question needs to be answered on Moodle, using the safe exam browser. This part of the exam is closed book and has a duration of 30 minutes.

Turn the page for question 2.

Question 2 **(35 marks)**

(a) The origin of the body frame B of a rigid body has coordinates $[\ -3.0 \quad 1.5 \quad 0.7\ ]^T$ in the inertial frame A. The positive directions of the x and y axes of frame B are in the positive y and z directions of frame A, respectively. The rigid body now moves with $\pi/6$ radians with the twist ξ_1, which is given to you in double precision in the Excel spreadsheet file "xi_q2_dat.xlsx", in the "Exam_B_files.zip" file on Moodle, under the "Exam" topic, or as indicated below:

$$\begin{bmatrix} 0.584897651865602 \\ -0.589396864572260 \\ 2.047141781529606 \\ 0.449921270665848 \\ 0.584897651865602 \\ 0.674881905998771 \end{bmatrix}$$

If you choose to read the data from the file into MATLAB, you can use the command

```
xi1 = readmatrix('xi_q2_dat.xlsx')
```

After this motion the body frame is re-defined as frame C. Calculate the homogeneous representation $g_{ac} \in SE(3)$. Explain in your script how you calculate your result, and report on the elements of g_{ac} both in your script and in the "Exam B Question 2 variables" feedback tool on Moodle. (On Moodle you need to report only on the 1st three rows of g_{ac}).

(b) The body moves a second time. After this motion the body frame is re-defined as frame D. The coordinates of the origin of D in frame A are $[\ 4.0 \quad 2.5 \quad -1.7\ ]^T$ and the directions of the axes of frame D with respect to the inertial frame is defined by the rotation matrix R_{ad}. The components of R_{ad} are given to you in double precision in the Excel spreadsheet file "Rad_q2_dat.xlsx", in the "Exam_B_files.zip" file on Moodle, under the "Exam" topic, or as indicated below:

$$\begin{bmatrix} 0.135526185435788 & 0.956172494898442 & -0.259551176188652 \\ -0.677630927178938 & 0.280570323320565 & 0.679776890017899 \\ 0.722806322324201 & 0.083752335319572 & 0.685956679927152 \end{bmatrix}$$

If you choose to read the data from the file into MATLAB, you can use the command

```
Rad = readmatrix('Rad_q2_dat.xlsx')
```

Calculate the rigid transformation matrix $g_{cd} \in SE(3)$ associated with the motion described above. Explain in your script how you calculate your result, and report on the elements of g_{cd} both in your script and in the "Exam B Question 2 variables" feedback tool on Moodle. (On Moodle you need to report only on the 1st three rows of g_{cd}).

Turn the page for question 3.

Question 3 (35 marks)

Consider the 3-DOF manipulator shown in figure 3.1 in the reference configuration. Link 1 of length l_1 is connected to the base through a revolute joint, joint 1, along axis ξ_1, in the positive z axis direction of the spatial frame S. In the reference configuration

- the centrelines of link 1, link 2 of length l_2 and link 3 of length l_3 are all aligned (concentric) and parallel to the y axis of S,
- the tool frame T origin is a distance $l_1 + l_2 + l_3$ away from the z axis of S (the tool frame itself is not indicated in the figure, just its origin), while the respective axes of T are parallel to the corresponding axes of S,
- link 2 is connected to link 1 through a revolute joint, joint 2, along axis ξ_2, in the negative x axis direction of S, and
- link 3 is connected to link 2 through a revolute joint, joint 3, along axis ξ_3, in the positive y axis direction of S.

Point p_b on intersection of the centreline of link 1 and ξ_1 has coordinates $[\ 0 \quad 0 \quad l_0\ ]^T$ in the spatial frame. Point q_w is at the location of joint 3 and has coordinates $[\ 0 \quad l_1 + l_2 \quad l_0\ ]^T$ when the manipulator is in the reference configuration.

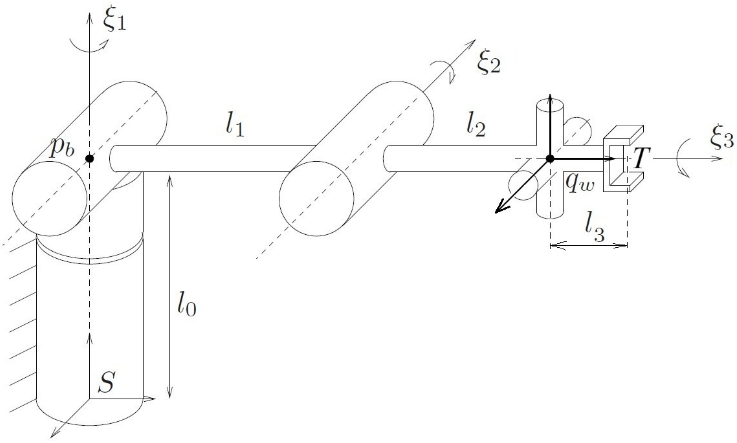

Figure 3.1

The lengths are $l_0 = 0.65$ m, $l_1 = 0.8$ m, $l_2 = 0.5$ m and $l_3 = 0.2$ m.

The manipulator is actuated and moves so that the configuration of the tool frame can be described by $g_{st} \in SE(3)$, given in components in the spatial frame, in double precision, in the Excel spreadsheet file "q3dat.xlsx", in the "Exam_B_files.zip" file on Moodle, under the "Exam" topic, or as indicated below in single precision (which may work in this case, but it depends on your calculation):

$$\begin{matrix} 0.77421181 & -0.41948725 & 0.47394780 & -0.75250222 \\ -0.15613988 & 0.59908988 & 0.78530991 & 1.07468455 \\ -0.61336482 & -0.68199836 & 0.39832377 & 0.17260115 \\ 0.00000000 & 0.00000000 & 0.00000000 & 1.00000000 \end{matrix}$$

If you prefer to read the data from the file into MATLAB (this is recommended), you can use the command

```
gst = readmatrix('q3dat.xlsx')
```

Calculate the joint angles θ_1, θ_2 and θ_3 through which joints 1, 2 and 3, respectively, rotated, in radians, to bring the tool frame to configuration g_{st} from the reference configuration. Explain in your script how you calculate your results, and report on these three angles both in your script and in the "Exam B Question 3 variables" feedback tool on Moodle.

Turn the page for question 4.

Question 4 (15 marks)

Consider the same 3-DOF manipulator that was studied in question 3, shown in figure 3.1, in the reference configuration. The reference configuration in this question is exactly the same as in question 3.

You are given an initialization MATLAB script file “MCandN_for_computed_torque_control.mlx” and a SIMULINK template file “q4_template.slx” in the “Exam_B_files.zip” file on Moodle, under the “Exam” topic. Once executed in MATLAB, the initialization script initializes all parameters necessary for the “MCN” function block in the SIMULINK template file to run successfully. You can accept that the initialization file and the “MCN” SIMULINK function block correctly calculates the $M(\theta)$, $C(\theta,\dot{\theta})$ and $N(\theta,\dot{\theta})$ matrices for the manipulator, as are used in equation (4.24) (p. 171) of the course textbook by Murray, Li and Sastry.

The SIMULINK template also contains a block that generates a rounded step function, which is the rounded step function that you should use in answering this question. This block has three outputs, labelled as “Displacement”, “Velocity” and “Acceleration”. These outputs are the rounded unit step, its first and its second time derivative, respectively.

You need to build working a SIMULINK model using the template, to simulate the manipulator under computed torque control, with rounded step input in the desired joint angle vector with step magnitude as follows: $\theta_d = [\ 0.5 \quad 1.0 \quad 0.8\ ]^T$. For the computed torque control scheme you should use the feedback gain matrices that are already calculated in the “MCandN_for_computed_torque_control.mlx” file.

Plot the response trajectory of the joint angle θ over a period of 4 seconds after the beginning moment of the step (which is set as 1 second in the initialization file). Plot all vector element trajectories on a single figure and properly identify in text or with a legend on the graph, which curve is which variable. You need to upload the figure to the “Exam B Question 4” assignment on Moodle, in `.jpg`, `.png` or `.pdf` format. You also need to write a brief explanatory description of each trace in your script, in which you interpret your result, but which will also serves to confirm that the figure uploaded to Moodle was in fact generated by yourself during the examination.